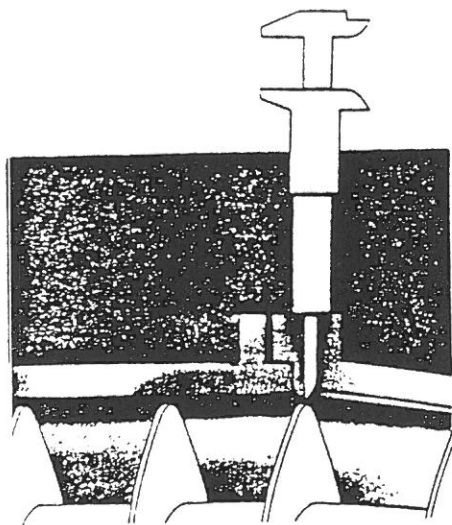
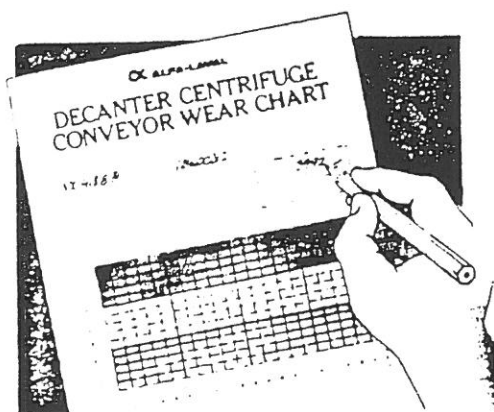


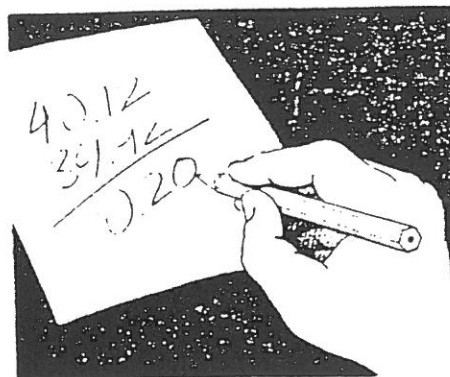


4. Insert a caliper gauge in the hole and measure the depth. Measure to the highest point on the edge of the flight.

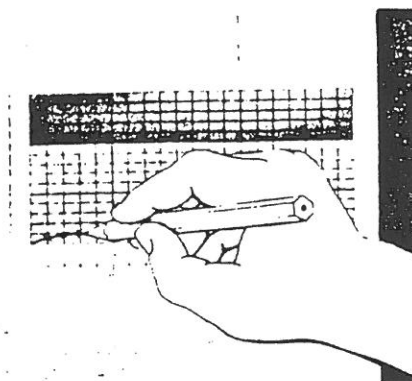
Please Note: The caliper gauge must always face in the same direction when measuring.



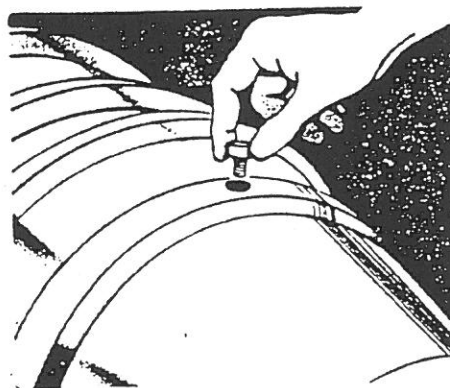
5. Make the first measurement when the conveyor is new, before it has been operated. This is your calibration measurement. Make a note of it on the wear chart on next page.



6. Repeat the procedure after every 100 hours' operation. To find the wear, subtract the original calibration reading from the latest reading.

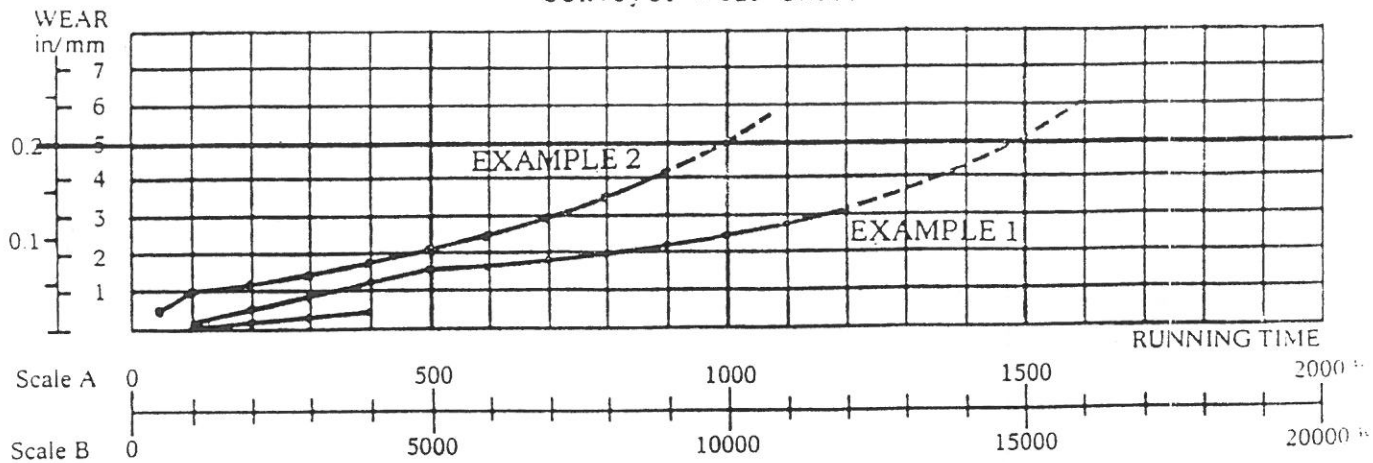


7. Plot the readings on the wear chart (see next page).



8. Do not forget to replace the plug! Lubricate it first with castor oil or ball bearing grease, and tighten to a torque of 22 Nm (2.2 kpm, 16 lb.ft).

Conveyor Wear Chart



How to use the chart:

1. Measure for calibration before the conveyor is run for the first time and enter the calibration reading in the box above the chart.
2. Consult your Alfa-Laval service representative about the acceptable limit of wear for the duty in question. Mark the limit by drawing a line across the chart (at 5 mm in example above).
3. Measure every 100 hours and plot the readings on the chart according to Scale A (0-2000 hours).
4. If wear is less than 1 mm (0.04") after 400 hours, shift to Scale B (0-20000 hours) and measure wear every 500 or 1000 hours.
5. By extrapolating from the plotted curve you will be able to predict approximately when the conveyor will need servicing.

Example 1:

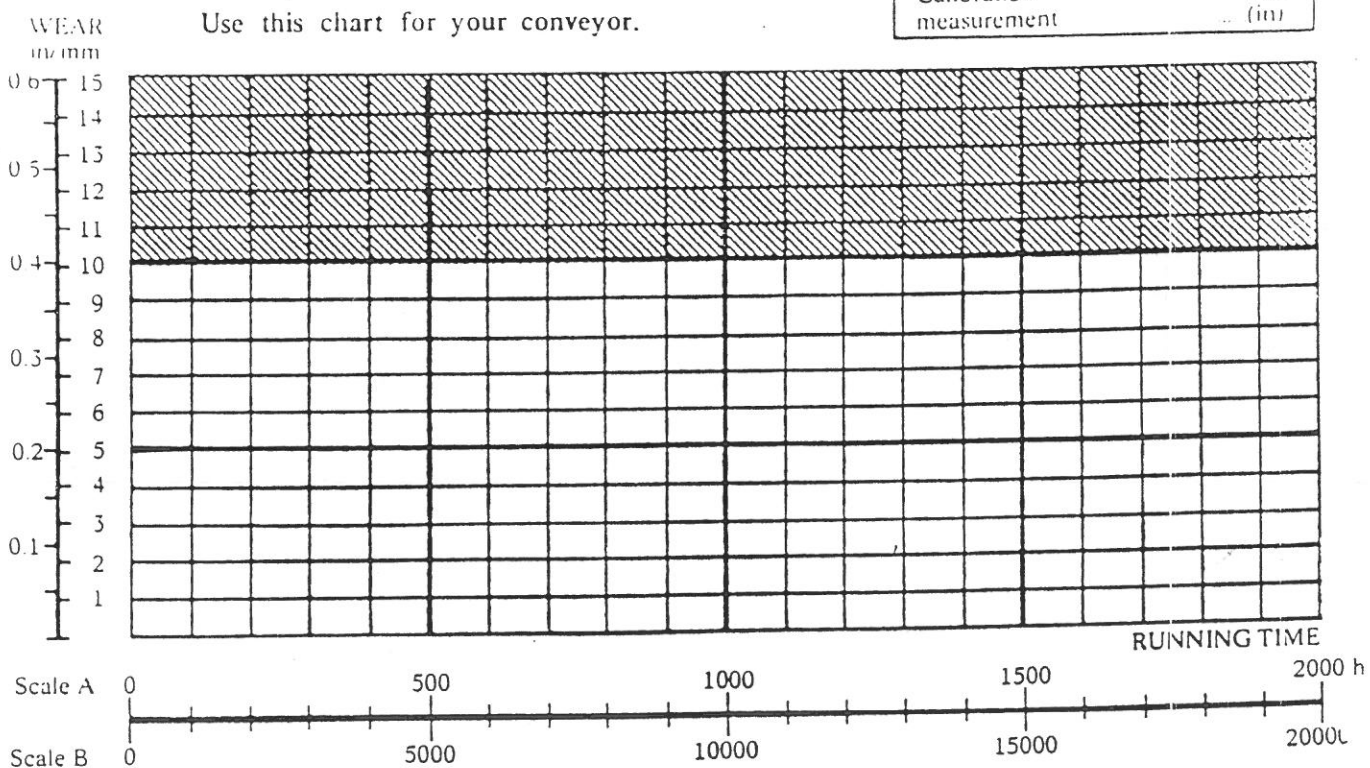
Wear is just over 1 mm (0.04") after 400 hours. Continue to measure every 100 hours. After 1200 hours, extrapolation indicates that the acceptable limit of wear (in this case 5 mm) will be reached at about 1500 hours.

Note the characteristic profile of the wear curve: first a linear rise until the conveyor is run in, then a period of stability and finally a progressive increase in the rate of wear.

Example 2:

Wear is only 0.5 mm (0.02") after 400 hours. The frequency of measurement can now be reduced to 1000 hours (Scale B).

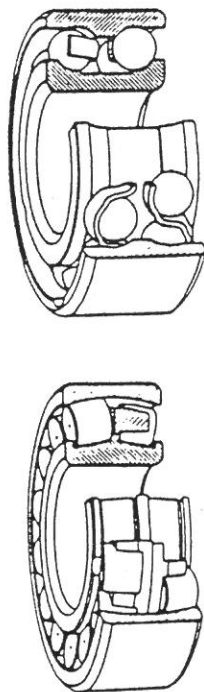
After 9000 hours, extrapolation indicates that the acceptable limit of wear will be reached at about 10000 hours.



Additional copies of Conveyor Wear Chart are forwarded on demand.

MAINTENANCE

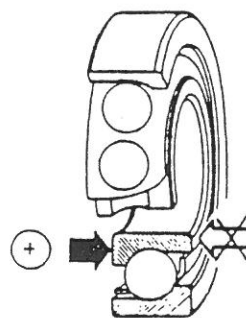
Bearings (ball bearings and roller bearings etc.):



- Make sure that no impurities (burrs) get into the bearing during mounting or afterwards. Never spin an ungreased bearing.

Never blow a bearing with compressed air.

Single-row Angular Contact Ball Bearings:



Note the following directions for inspection and exchange of bearings:

- Leave the bearing in its wrapping until ready to fit.
- A new bearing is covered with grease. Leave the grease untouched.
- If a bearing is heated in oil before mounting, the oil temperature must not exceed 80°C (175°F).
As soon as the bearing is thoroughly warm, remove it from the oil bath.
- A bearing should never be forced on to a shaft by blows applied to the outer race, nor into a housing or seat by pounding on the inner race.
- A bearing which has been taken out of the machine must be replaced.

In a single-row angular contact ball bearing each ball race has a high and a low shoulder. It is essential to fit the bearing the right way round, as an axial load on the low shoulder would damage the bearing.

In SKF bearings an impressed mark (+) indicates the side of the inner ball race on which the axial load should act.

V-belts: New belts can be fitted only after the feed pipe has been removed.

Move the motor on its slide rails, pushing it towards the machine so that the belts can be placed in the pulley grooves by hand. **Never** force the belts into place with tools or the like.

Align the motor belt pulley with the machine belt pulley.

Check that the motor is not pulled askew and that the belt pulleys are in line with each other so that the belts run straight.

Tighten the belts as described on the next page.

Fasten the belt guard and the feed pipe.

Belt Tension: It is important that the belts are correctly tensioned. If a belt is too loose it will slip on its pulley and quickly wear out; if it is too tight it will cause wear on the bearings.

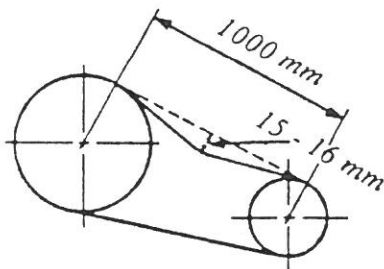
Proceed as follows to get the correct tension:

1. Take up the slack at the top. Adjust the distance between centres to a light tension and rotate the drive parts a few turns by hand.

2. Adjust the distance between centres until all belts lie well.

Start the motor and run the drive parts for some minutes so that the belts can seat themselves in the grooves.

3. Stop the motor and adjust the distance between centres until all belts lie evenly and are properly tightened.



*Depression between pulleys:
15 - 16 mm per 1000 mm distance between
shaft centres. (5/8" per 40" shaft distance).*

In a new machine the belts should be retensioned after some time. After this, further tightening will mostly be unnecessary, as the length of the belts remains practically constant. When exchanging belts, never force them over the side of the pulley, but always loosen the motor mounting first.

Check the belt tension from time to time.

Cleaning: The working life and proper functioning of the belts depend to a high degree on how they are cared for.

Oil and grease will soon perish and ruin them.

Oily or greasy belts can be suitably cleaned with a rag soaked in some grease-dissolving agent such as petrol or trichloroethylene.

Exchange: When one or more of the belts are worn out, put on a complete new set of belts.

Storage: If the machine is taken out of operation for some time, the belts should be removed from the pulleys and stored hanging - not rolled up - in a cool, dark place.

Spare belts in sets should be stored in the same way.

Belt Pulley with "TAPER-LOCK" Bushing:

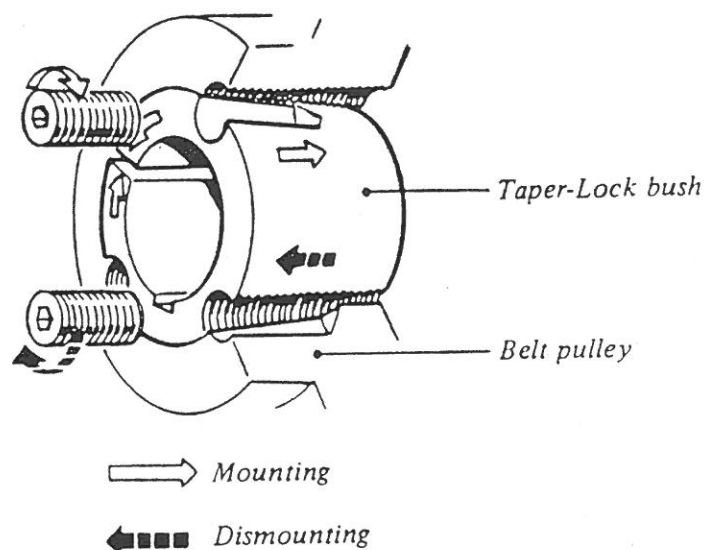
The sketch on next page shows the hub of a belt pulley fitted with a "Taper-Lock" bushing.

The split bush is cylindrical on the inside and tapered on the outside, in which there are two semi-cylindrical dead-end holes (in larger bushes three) and one semi-circular threaded through hole.

The bore in the belt pulley hub has the same taper as the clamping bush. In the hub bore, facing the dead-end holes of the bush, are two semi-cylindrical threaded through holes and, facing the threaded hole of the bush, a semi-cylindrical dead-end hole.

When mounted, screws placed in the threaded bores of the hub will press the clamping bush into the tapered bore of the hub. Being split, the bush will clasp the shaft with increasing force as the screws are tightened. When the bush is pressed far enough into the hub bore, it will clamp the shaft with the same force as if mounted with an interference fit.

Belt Pulley with "TAPER-LOCK" Bushing



Mounting (upper half of figure):

Carefully clean the parts.

Put the bush into the hub and align the holes.

Place the screws in the threaded holes of the belt pulley hub.

Push the unit on to the shaft and align it axially.

Tighten the screws alternately, using a hexagon key. Tighten until the Taper-Lock bush is pressed in far enough.

Dismounting (lower half of figure):

Remove the screws and place one of them in the threaded hole of the Taper-Lock bush.

Tighten the screw until the bush comes loose.

The belt pulley can be pulled off by hand.

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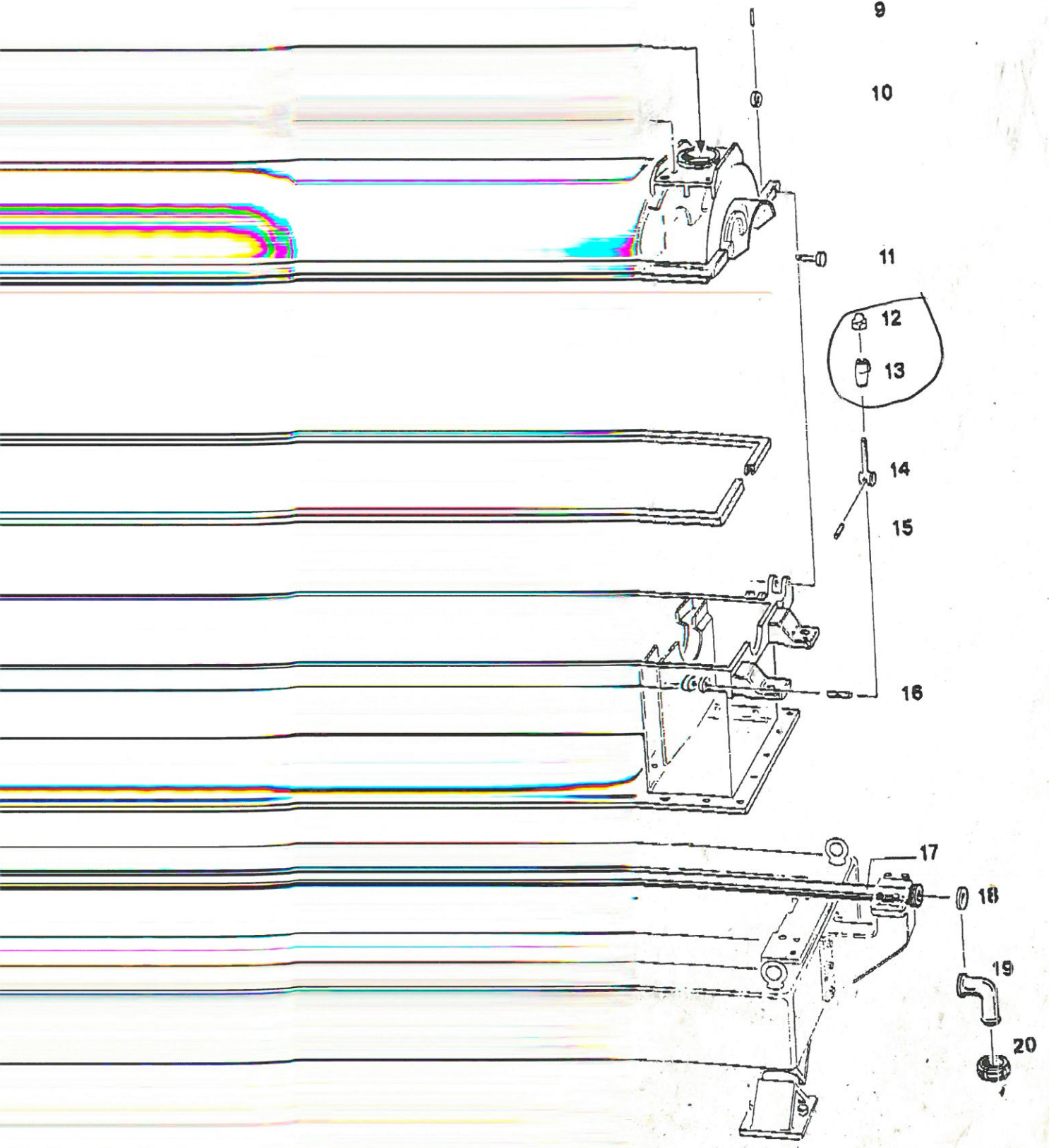
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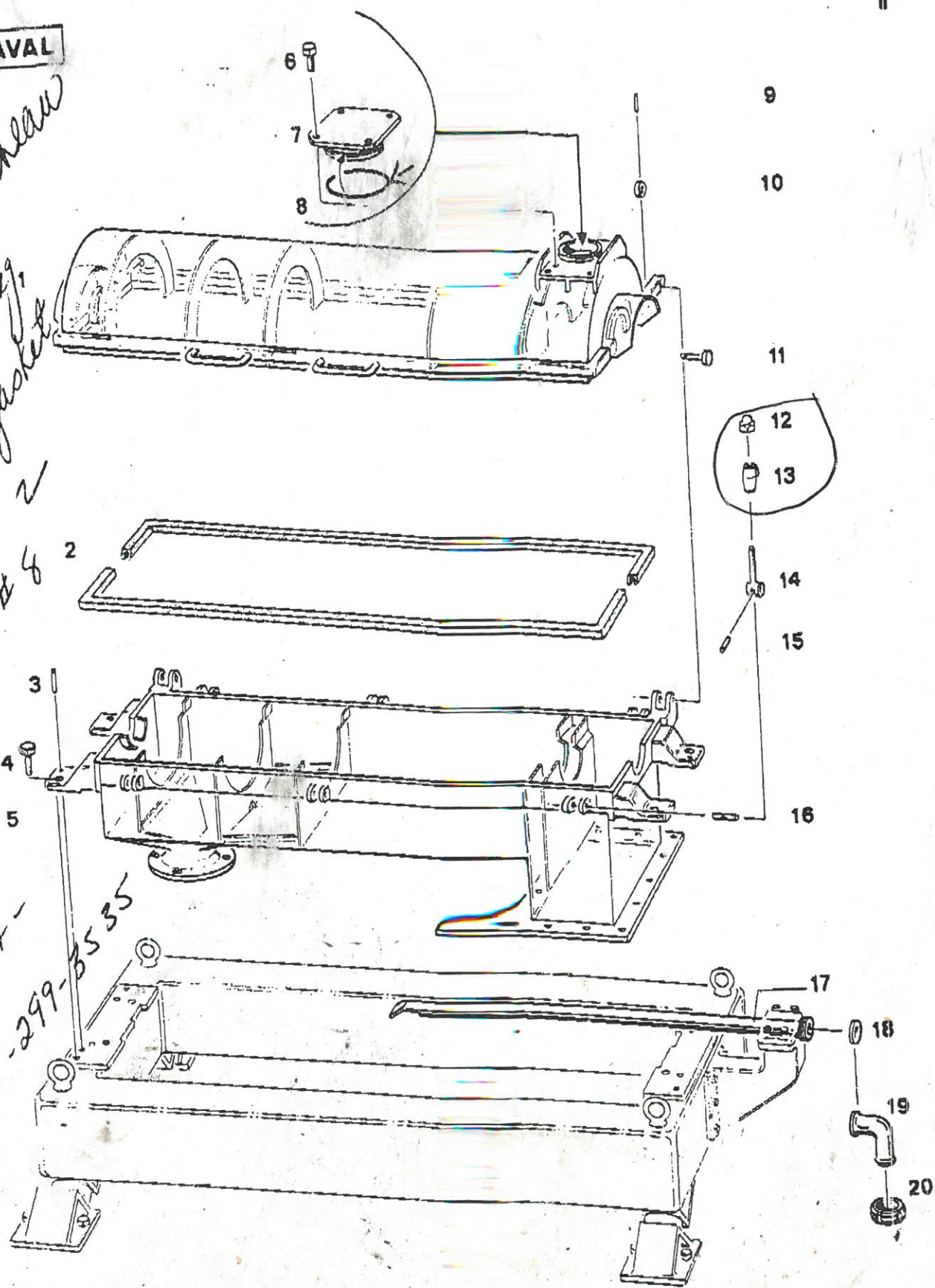


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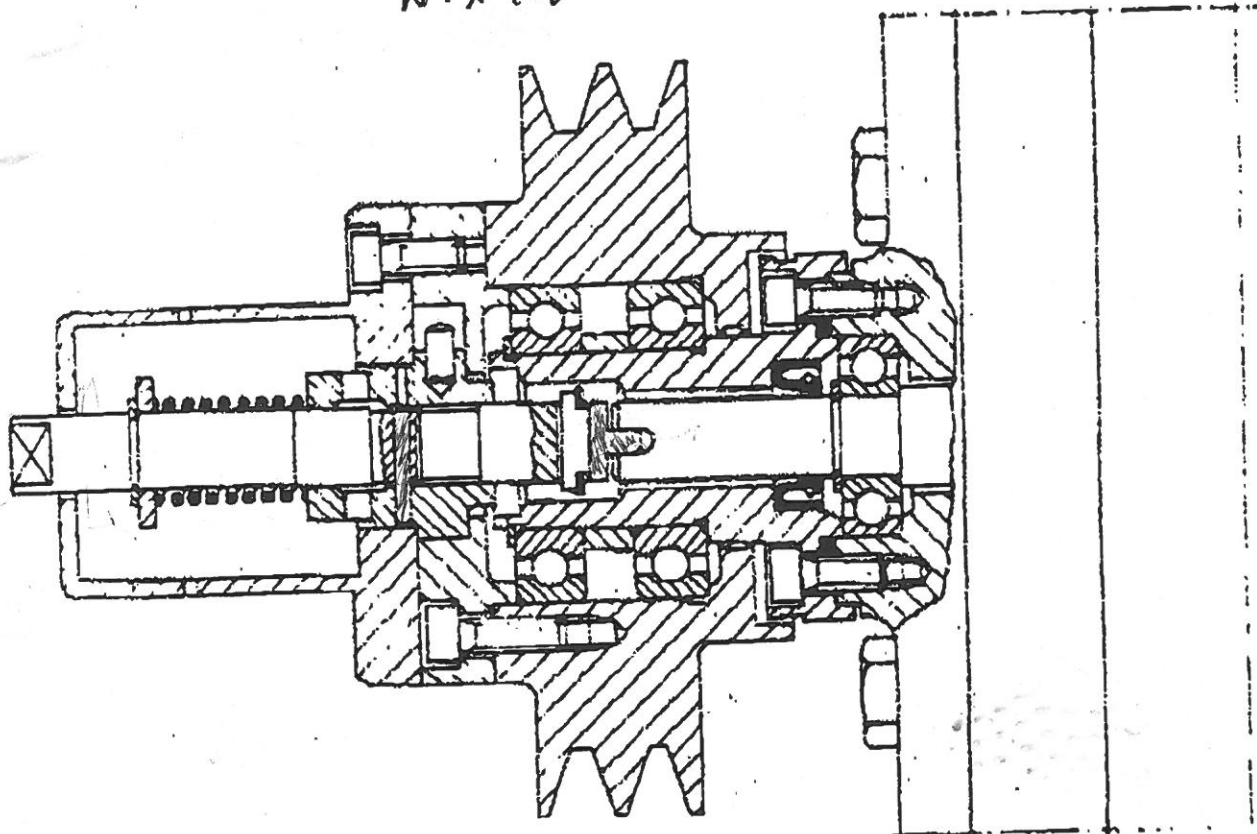
ALFA-LAVAL

GEAR BOX

P

GEAR BOX with SHEAR PIN COUPLING and DEVICE
FOR DRIVING the SMALL SUN WHEEL

G.S. TYPE COUPLING ON CITY ST THOMAS
N.X 418



Part number	Complete gear box with shear pin coupling and device for driving the small sun wheel
518032-86	Comprising: Gear box No. 518032-82 excl. of packing No. 66220 and sealing cover No. 66219. Unit comprising shear pin coupling and device for driving the small sun wheel -- 518032-87. Parts -- see next page.

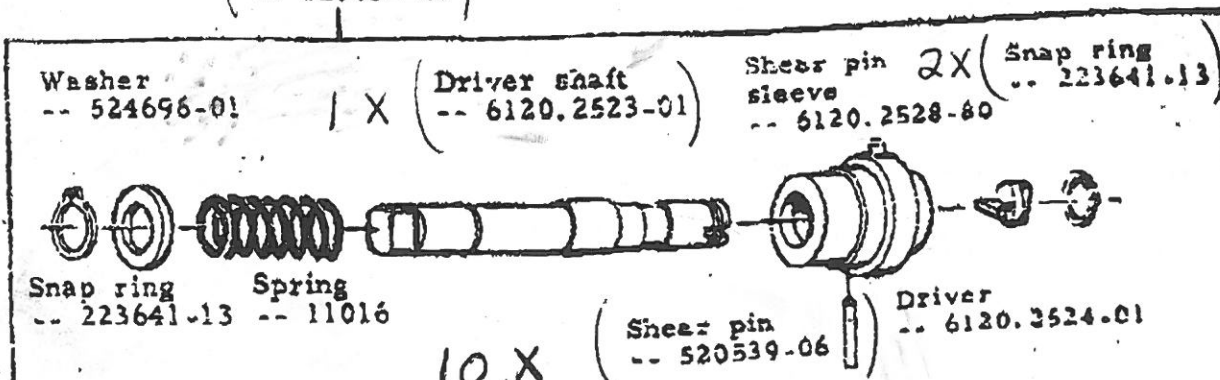
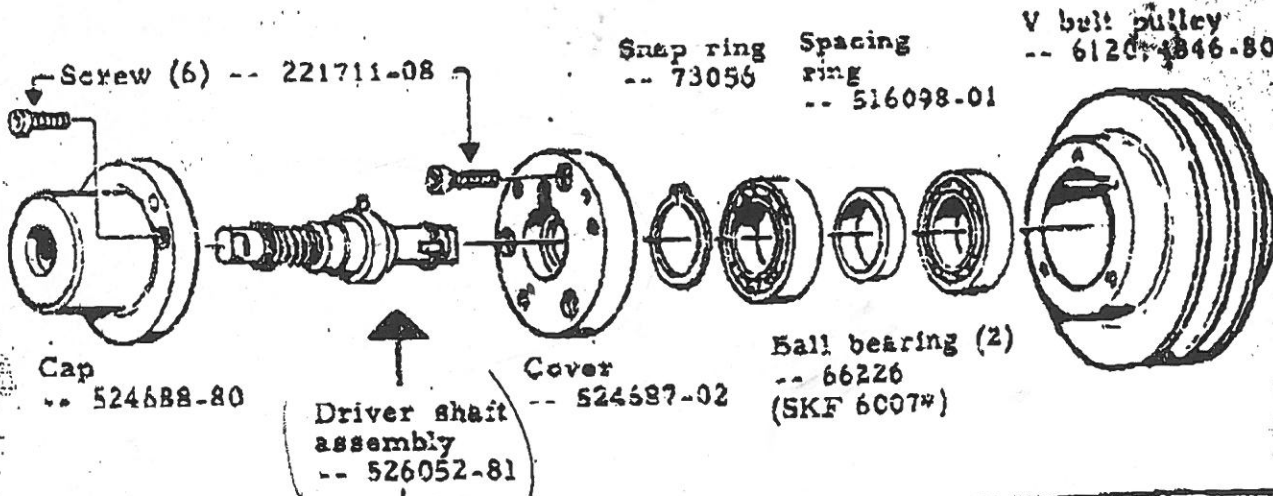
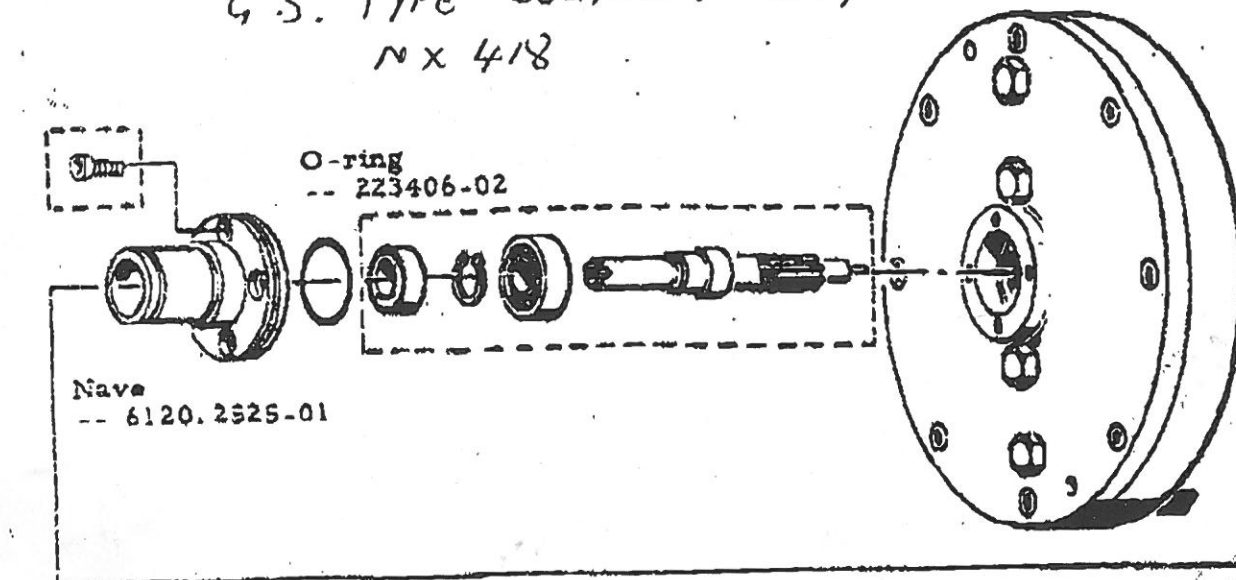
ALFA-LAVAL

GEAR BOX

P

SHEAR PIN COUPLING and DEVICE FOR
DRIVING the SMALL SUN WHEEL -- 518032-87

G.S. TYPE COUPLING CITY ST THOMAS
N X 418



or equivalent bearing of another make.

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TIGHTENING OF SCREWS

Tightening all screws with the right torque is important.
The table indicates the torques for metric and UNC thread.

Metric Thread			
mm	Quality Class 8.8 TORQUE		
	kpm	Nm	lb.ft
3	0.12	1.2	0.87
4	0.28	2.8	2.02
5	0.56	5.6	4.05
6	0.95	9.5	6.87
7	1.57	15.7	11.35
8	2.29	22.9	16.56
9	3.35	33.5	24.22
10	4.53	45.3	32.75
12	7.86	78.6	56.83
14	12.60	126.0	91.10
16	19.20	192.0	138.80
18	26.80	268.0	193.80
20	37.50	375.0	271.10
22	50.70	507.0	366.60
24	64.80	648.0	468.50

UNC Thread			
Inch	Quality Class D 80 TORQUE		
	kpm	Nm	lb.ft
1/4"	1.17	11.7	8.46
5/16"	2.37	23.7	17.14
3/8"	4.18	41.8	30.22
7/16"	6.67	66.7	48.22
1/2"	10.10	101.0	73.00
9/16"	14.40	144.0	104.10
5/8"	19.90	199.0	143.90
3/4"	35.00	350.0	253.10
7/8"	51.30	513.0	370.90
1"	76.80	768.0	555.30

VIBRATION SENSOR

The Vibration Sensor (optional extra equipment):

The machine can be equipped with a vibration sensor to protect it against damage due to heavy vibration.

The vibration sensor - the vibraswitch - protects the machine by cutting off the power supply to the decanter motor and the feed pump if the decanter begins to vibrate excessively. The control panel - or the motor starter - must be provided with a terminal for the vibraswitch.

Highest permissible vibration level = 1.0 g above the normal vibration level.

Being erected on vibration dampers, the decanter deflects greatly during starting and stopping, but these oscillations are so soft that they will not influence the vibraswitch.

The vibraswitch is reset manually by pressing the reset button - see sketch below. (The built-in reset coil of the sensor not to be connected).

Adjusting:

First remove the cover of the vibraswitch.

1. Zero vibration point of sensor with non-operating machine. The adjustment is made with the machine not operating. Turn the adjusting screw counter-clockwise two turns and press the reset button.

Then turn the adjusting screw slowly clockwise until the armature trips and actuates the switch. This is the zero vibration point, or actuating point, of the sensor with the machine not operating.

Mark out this position with a line right before the adjusting screw pointer.

2. Normal vibration level with operating machine. The adjustment must be made during normal operation. Turn the adjusting screw - see sketch below - counter-clockwise one turn and press the reset button. If now the vibra-

switch is not retained in the reset position, turn the adjusting screw counter-clockwise another turn or two, so that the vibraswitch remains in the reset position when the reset button is pressed.

Then turn the adjusting screw slowly clockwise until the armature trips. Mark out this position with a line right before the adjusting screw pointer.

The difference between the two actuating points in steps 1 and 2 is the normal vibration level during operation expressed in scale divisions. One scale division, i.e. the distance between two scale lines, is 0.1 g, one full turn is 1.0 g.

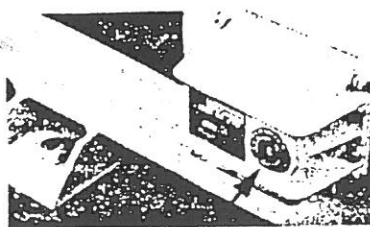
3. Setting of highest permissible vibration level. The adjustment must be made during operation.

The highest permissible vibration level should be 1.0 g greater than the normal vibration level during operation. Thus, turn the adjusting screw counter-clockwise one full turn from the point where actuation occurs according to step 2.

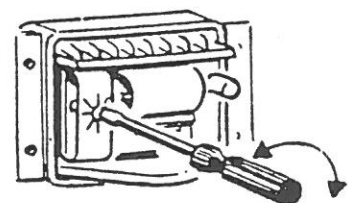
Finally, put on the cover and fasten it with the screws.

Note! Readjustment may be required if the vibraswitch shuts off the power supply because the building begins to vibrate considerably when the machine is operating.

Quantity	Part	Article No.
1	Vibraswitch model 366	53.0307-01
4	Screws	6119.4100-67
4	Washers	6119.4400-06
4	Spring washers	6119.4430-06



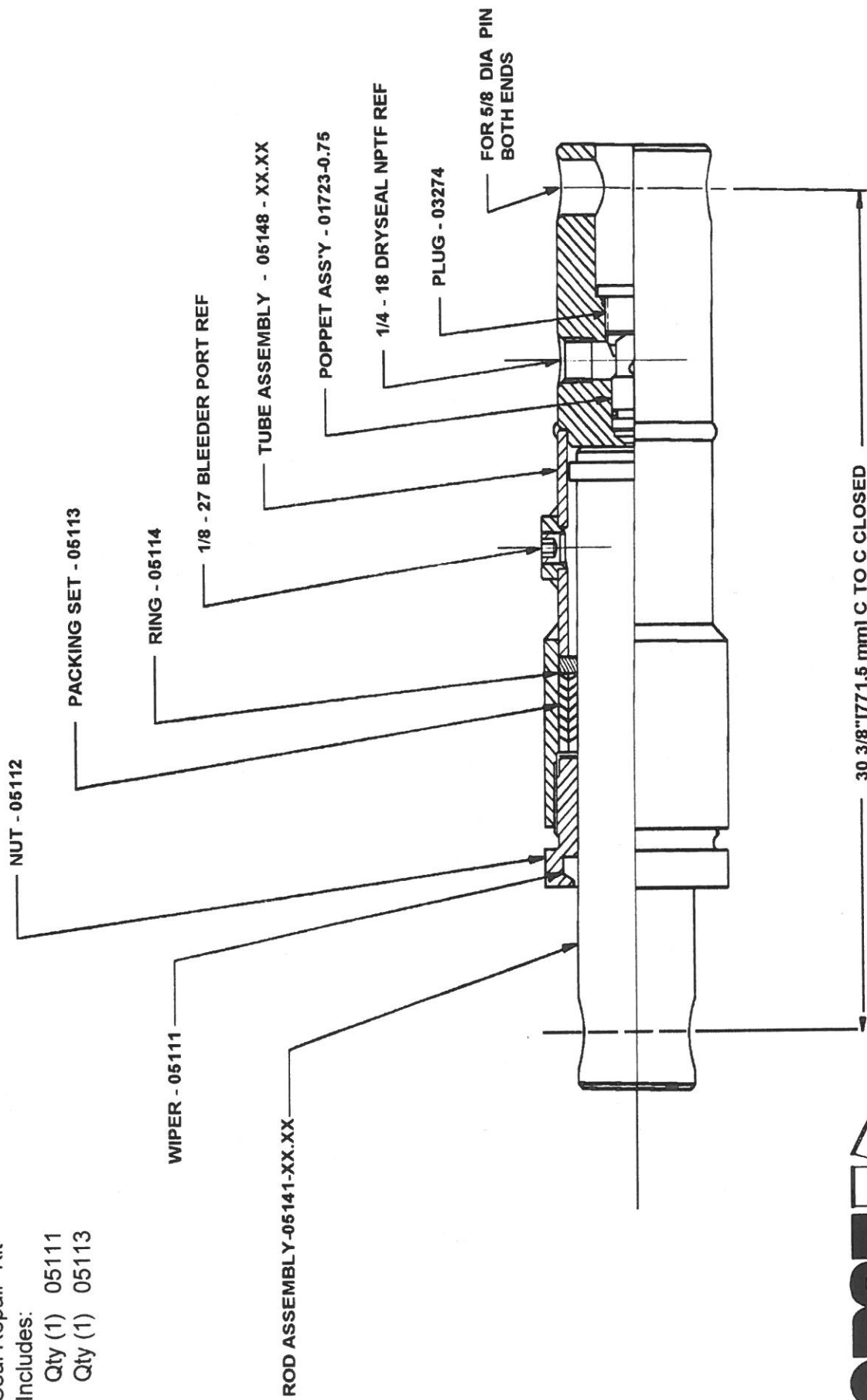
Reset button



Adjusting screw

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Qty (1) 05113



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